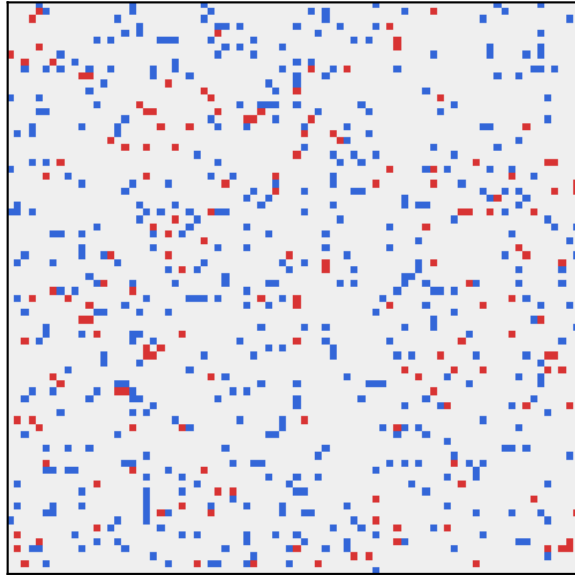


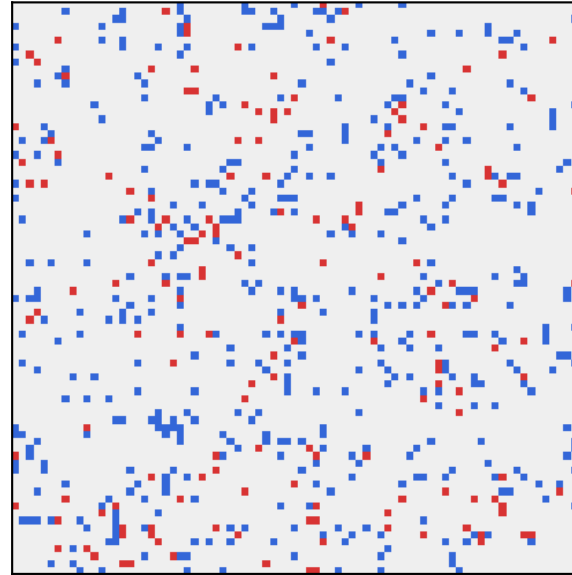
CFAC-compliant bias variants: Hill saturation $w \propto 1 + \chi_m \phi / (K + \phi)$ keeps (I1) and can produce spatial concentration

No bias (canonical)



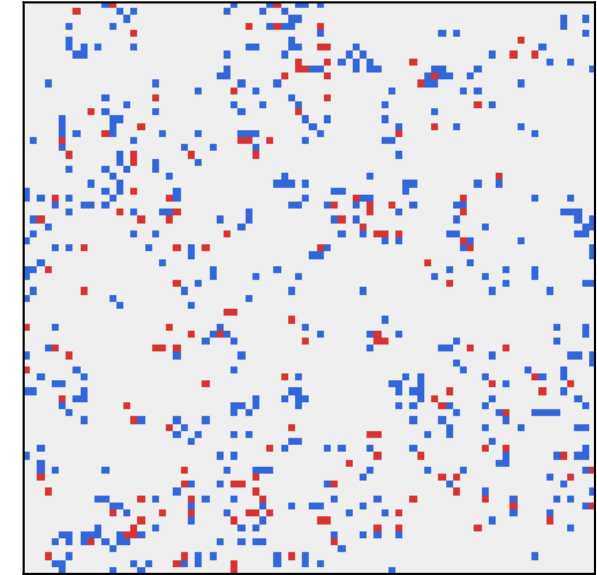
$\langle \phi \rangle = 0.47$ max=12.0

Moderate Hill bias



$\langle \phi \rangle = 0.46$ max=11.2

Strong Hill bias
(saturating, still bounded)



$\langle \phi \rangle = 0.47$ max=11.0

Pheromone ϕ

